# Unit 1

## INTRODUCTION TO SEARCH ENGINE

# Mission of search engines

- Deliver Relevant Results
  - The primary mission is to provide the most relevant and high-quality search results in response to a user's query quickly and accurately, ensuring user satisfaction.
- Organize information
  - Search engines systematically crawl, index, and organize the vast amount of data on the internet to make it easily searchable and accessible for users.
- Provide fast and fresh responses
  - They aim to deliver search results as fast as possible while keeping the information fresh and up-to-date, facilitating a better user experience.
- Combat spam and ensure quality
  - Search engines continuously work to prevent spam and manipulate content to maintain the quality and trustworthiness of search results.
- Enhance user experience and satisfaction
  - They personalize and refine search results based on user behavior, location, and preferences to increase the usefulness and relevance

# Market share of search engines

- Google dominates the global search engine market with a commanding share of about 89-90%, making it the clear leader worldwide across all devices.
- Microsoft's Bing holds the second position but with a significantly smaller share, around 4-4.1% globally.
- Yandex leads the market in Russia with about 76.3% share locally, though globally it holds around 1.65-2.5%, reflecting strong regional preference.
- Other search engines like Yahoo!, DuckDuckGo, and Baidu have smaller global shares roughly between 0.6% and 1.5%

# Human goals of searching

- Information seeking
  - People search primarily to acquire knowledge or answers to specific questions
- Decision making
  - Search helps users gather data to make informed decisions, whether for research, shopping, or solving problems in daily life or work.
- Entertainment
  - Some searches are motivated by the desire for entertainment, such as watching videos, exploring hobbies, or browsing interesting content
- Dependency and external memory
  - Humans use search engines as external memory aids to quickly retrieve information, reducing the mental effort to remember facts or details.
- Exploration and learning
  - Searching enables users to browse, explore, and learn progressively about topics, supporting cognitive growth and long-term knowledge development.

# Determining Searcher Intent:

## A Challenge for Both Marketers and Search Engines How People Search

- Ambiguity of User Queries
  - Search terms are often ambiguous and can mean different things to different users.
- Types of Search Intent
  - User searches generally fall into categories such as informational (seeking knowledge), navigational (finding a specific site), and transactional (wanting to take action, like buying).
- SERP and Keyword Analysis
  - Analyzing search engine results pages (SERPs), keyword modifiers ("how to," "buy," "review") and query language helps decode what users want
- Overlapping and Evolving Intents
  - A single search may combine multiple goals, such as research and purchase, and intents can shift quickly.
- Personalization and context
  - Marketers and engines factor in user location, device, history, and context to refine results, but balancing privacy and precision remains a challenge

# How Search Engines Drive Commerce on the Web

1. Product Discovery
   - Search engines enable users to quickly find and research products and services, helping them make informed purchase decisions
2. High conversion rates
   - Visitors using site search are 2–3 times more likely to convert, with product search features increasing overall sales
3. Mobile commerce impact
   - Over half (65%) of “I-want-to-buy” searches happen on mobile devices, showing the importance of mobile-optimized search for online sales.

4. Organic traffic power
   - Organic search delivers 53% of all website traffic for many online stores, ensuring sustained growth and higher conversion rates
5. Revenue generation
   - Consumers who find products via search tend to spend more, with organic search often driving as much as 14–45% of total e-commerce revenue
6. Business insights
   - Search engine analytics provide vital insights into consumer behavior and preferences, helping businesses improve their marketing strategies.
7. Enhance personalization
   - Search engines tailor results with personalized recommendations and autocomplete, boosting sales by up to 24% and increasing users’ likelihood to return and buy again.

# Eye Tracking: How Users Scan Results Pages

- Eye tracking studies reveal that users typically scan search engine results pages (SERPs) in distinct and repetitive patterns, with the most prominent being the F-pattern and the Golden Triangle.
- Most attention is focused at the top-left corner of the results page, capturing the first few results most extensively.
- Users scan down the page vertically through 2–3 results, then horizontally across—forming an “F-shape” pattern with their eye movements.
- Lower results receive significantly less attention, emphasizing the importance of top rankings in organic search.
- Bold keywords, titles, and descriptions further attract users’ attention and dictate their scanning behavior.

## SAMPLE DIAGRAM

```
|SEARCH QUERY|
| Result 1: ************** |
| Result 2: *********     |
| Result 3: *****        |
| Result 4: **        |
| Result 5:       |
|-------------------------|
```

# Click Tracking: How Users Click on Results

Users interact with search result pages in distinct ways, and tracking how they click on results provides deep insights into search behavior and website optimization.

- Most clicks happen rapidly: About 50% of users click on a search result within 9 seconds, and the average time to first click is approximately 14.6 seconds
- Autocomplete suggestions matter: Around 23% of users choose one of Google's autocomplete suggestions
- Organic results get the majority of clicks: Approximately 65% of users end up clicking an organic result

CONT….

- Ads attract clicks, but less frequently: Nearly 19% of users click on a Google ad
- Most users focus on the top results: Over 80% of first clicks go to the top four results. The first page dominates user attention
- Few users explore beyond page one: Around 75% to 90% of searchers never go to the second page,

# SEO GOALS AND OBJECTIVES

- Increase Organic Traffic
  - The primary SEO goal is to grow the number of unpaid, relevant visitors from search engines to your website to drive sustainable, cost-effective traffic.
- Improve Keyword Rankings
  - Achieve higher positions for targeted high-value keywords in search engine results pages (SERPs). Better rankings increase visibility and click-through rates, directly impacting organic traffic and conversions.
- Enhance Content Quality and Relevance
  - Develop comprehensive, valuable, and up-to-date content that matches user search intent.

CONT…..

- Fix Technical SEO Issues
  - Ensure the website is free from crawl errors, slow speeds, mobile incompatibility, indexing problems, and other technical barriers. A technically sound site is crucial for search engines to crawl, index, and rank pages effectively.
- Boost User Experience (UX)
  - Optimize website navigation, page load speed, mobile usability, and overall interaction to reduce bounce rates and lengthen session duration
- Acquire High-Quality Backlinks
  - Build authoritative and relevant inbound links through ethical strategies like guest posting, partnerships, and link-worthy content creation.
- Increase Conversion Rates from Organic Traffic
  - Focus on converting search engine visitors into customers, leads, or desired actions by optimizing landing pages, calls to action, and relevant offers.

UNIT 1
OVER